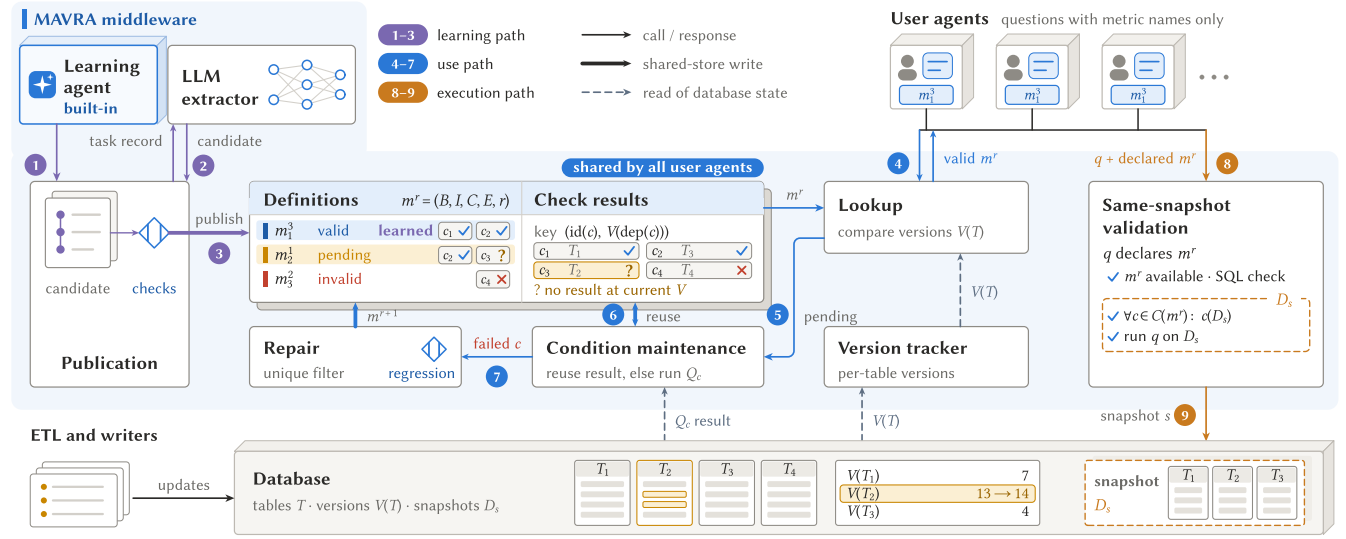

MAVRA middleware
Learning agent
built-in
LLM extractor
task record
candidate
1–3 learning path
4–7 use path
8–9 execution path
call / response
shared-store write
read of database state
User agents
questions with metric names only
shared by all user agents
Definitions
$m^r = (B, I, C, E, r)$
Check results
valid
learned
pending
invalid
key (id(c), V(dep(c)))
? no result at current V
publish
candidate
checks
Publication
Lookup
compare versions V(T)
valid m^r
q + declared m^r
Same-snapshot validation
q declares m^r
m^r available · SQL check
$\forall c \in C(m^r)$: $c(D_s)$
run q on D_s
reuse
pending
$V(T)$
Repair
unique filter
regression
failed c
Condition maintenance
reuse result, else run Q_c
Version tracker
per-table versions
Q_c result
snapshot s
ETL and writers
updates
Database
tables T · versions $V(T)$ · snapshots D_s
$V(T_1)$ 7
$V(T_2)$ 13 → 14
$V(T_3)$ 4
snapshot D_s